Arduino UNO V2.0 Design and learnings and Power path diode learning (V2.1)

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(Little grammatical refinement assisted by AI)

***Most important learning and lesson at the end.**

I have made is Arduino uno clone using ATmega328p and CH340G module. It is the second iteration of the same board, first iteration was earlier posted on LinkedIn, but was not documented. The first iteration was made on a zero PCB and was unsuccessful. Although decoupling capacitors and crystal were placed near to MCU, the board didn't initialize. After debugging it, the main reasons of failure were recorded as follows:

1. A basic physical layout mistake: I used solder wire blobs instead of short copper wiring for almost all connections. This resulted poor return path that defeated the purpose of close decoupling.
2. Although the schematic was nearly full correct, the PCB layout was not that great, as it had many long return paths.

So, because of above reason, specially because of the second one. I didn't order PCB fabrication from JLCPCB and decided to iterate a new version.

So now I made the second iteration of the uno board. With improved schematic and a far better PCB layout, thanks to previous learning mistakes and using a very useful tool 'via' and 'grounding using copper pour'. The following things were changed in schematic:

1. A diode was intentionally not placed before the7805 to preserve input headroom and avoid unnecessary dropout, especially for battery powered operation.
2. At the power output from 7805, I divided the power supply into two parts, one with 1N4007 diode to CH340G module (to prevent reverse charging from Arduino board) to the device and not sort the ports of device (also using the fact that the CH340G module can run from 3.3V-5V). And other without diode as a 5V main power supply to the MCU.
3. Not using any resistor at the DTR, RX, TX receiving to MCU in order to keep signal edges clean and because no EMI or protection constraint existed in this design.
4. Using two separate 100nF decoupling capacitors, for AVCC and VCC, cause although being internally connected to supply 5V to analog signal, both have separate noises.

The following are changes in PCB layout:

1. Common ground pour as a first priority (I have done on top layer). As gives a low impedance return path, which results in a stable MCU startup.
2. Keep components close and use short connects to reduce impedance.

3. Use of ‘via’ tool, as top jump between layers.
4. Special attention to closeness of crystal, decoupling capacitors and 10K reset.
5. Ensuring enough spacing between pads, routing and via holes.

Now moving into what things, I have learned in the process (electronics):

1. Decoupling capacitors: first of all, that it is just a normal capacitor used for a specific purpose. No matter how perfect supply you make, there will be some impedance and resistor in it which will cause voltage jumps and dips or spikes called noise. These noises highly effect the component it is attached to especially to an MCU. When placed near to MCU it provides a local charge reservoir to supply instantaneous transient current demands and reduce supply voltage fluctuations.
   Generally, for the purpose 100nF ceramic capacitor is used, as it has a low ESR and impedance. And when used near a power supply, generally used along with 10uF electrolytic capacitor. As 100nF handles high frequency transients, while the 10uF stabilizes lower frequency supply variations.
   And lastly know why decoupling is necessary, as without it, MCU can fail without any reason, and we will never know the reason of so.
2. RESET: the RESET pin of ATmega328p is active low and is held high using a 10K pull up resistor to ensure a stable operation. Meaning that reset condition is triggered when the pin voltage is pulled to logic 0 instead of logic 1. This forces MCU to go into a known initial state (restarts it). This pull up provides noise immunity, limits current during reset events, and allow safe external reset sources such as push button (here RESET button) and USB-UART DTR signals (minimum external circuitry).
3. Clock and crystal: the clock defines instruction timing, UART baud rate, controls timers, delays, PWM etc., and is set when the MCU initiates. Atmega328p, despite having an internal RC oscillator of 8Mhz, it uses an external 16Mhz crystal (and no we can not change crystal’s value, unless everything is recompiled from basics, cause all Arduino core libraries assume 16Mhz.
   And here we use a CRYSTAL OSCILLATOR CICUIT called Pierce oscillator, made up of a crystal (here 16Mhz) and two load ceramic capacitors (here 22pF, from data sheet). And here we have also used a 1M ohm resistor to provide DC bias and helping oscillator start reliably and prevent latch up during start up.
   And at last, using bootloader burning fuse bits set to an external 16Mhz crystal oscillator.
4. Bootloader burning: BOOTLOADER BURNING WILL BE EXPLORED ONCE THE FABRICATED BOARD ARRIVES AND PROCESS IS PRACTICALLY PERFORMED.

Now moving into what things, I have learned in the process (components and physical limitations):

1. Schematic correctness does not guarantee a working system (iteration1).
2. The routing length constraint becomes significant when using Through Hole components.
3. Return path quality is more important than proximity alone.
4. Of course, decoupling, crystal oscillator and reset should be placed closely with smallest possible traces.
5. Avoid using acute angle and right-angle bends while routing (causes speed issues). And avoid auto route at all costs.
6. PCB layout decision effects overall power integrity and thus working of system.
7. High current, voltage supplies be routed with thicker traces.
8. Use standard ICSP connections, unless firmware or pin mapping changes are intentionally made.
9. While working on zero PCB, use thin copper wire and never use solder blobs as a conductor.
10. Sometimes standard circuits must be adapted to real constraint (here not using diode at input of 7805).
11. AND MOST IMPORTANTLY DEBUGGING FAILURES TEACHES MORE THAN SUCCESSFUL BUILDS.

Next is now moving into iteration of nano clone using CH340G IC, instead of module to learn about tighter power integrity, surface mount components and how things change on moving from a single MCU on layout to an MCU and UART chip (USB-to Serial adapter chip).

From all possible cross-checks, this iteration appears both schematically and layout-wise improved, and is expected to function correctly after assembly and bootloader programming, after receiving from fabrications via JLCPCB. It will be a bit delayed as I will be ordering it along with nano clone and one more project (will be named after nano) all together to consider shipping charges.

NOW WHILE DESIGNING NANO CLONE, I REALIZED THAT I HAVE USED DIODE ORIENTATION IN MY V2.0 (FOR ME MOST IMPOTANT LEARNING), full explanation as follows:

As I am using CH340G IC instead of its module, everything matters, then suddenly I remembered that in my V2.0, power supply cannot be delivered to ATmega328P when using only USB input. ***AND MOST IMPORTANTLY, AS USAULLY I WILL BE USING AN***

<u>EXTERNAL POWER SUPPLY, AND WHEN I HAVE TO UPLOAD CODE THEN TO CH340G MODULE 5V FROM TWO DIFFENT SOURCES WILL GO WHICH WILL DAMAGE IT. AND IF I TRY TO AVOID IT, BY NOT USING EXTERNAL POWER SUPPLY WHILE UPLOADING CODE, THEN ATmega328P WILL NOT GET POWER CAUSE OLD DIODE ORINTATION WILL NOT ALLOW SO.</u>

<u>(AND FOR SURE ONE MORE REASON FOR FALIURE OF ZERO PCB ITERATION).</u>

As thanks to working on nano on right time, and always a habit of double checking and asking why this and that, I CORRECTED IT IN V2.1, which is now the final version. In V2.1 I cleaned layout even more by placing the diode near 7805.

So now what will happen is:

1. USB only: USB (5v, from CH340G module) to diode (here voltage drop) to Arduino.
   NOTE: LATER THE 1N4007(V2.0) IS ALSO REPLACED WITH 1N5819 TO REDUCE VOLTAGE DROP BY ~0.2-0.3V.
2. External only (when code is already uploaded): 7805 to Arduino, diode blocks back feed.
3. USB + external: Arduino powered by 7805, USB isolated. As it is physically impossible that 4.7v source push current into a node already at 5v.
4. CH340G is fully protected, and board will work as normal standard UNO board.